Complex Analysis Homework 7

张溢致

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19. Prove that the complex plane slit along the union of the rays $\bigcup_{k=1}^{n}\{A_k + iy : y \le 0\}$ is simply connected.

[Hint: Given a curve, first "raise" it so that it is completely contained in the upper half-plane.]

证明 任取该割开区域内的一条闭曲线 γ。取充分大的 $C > 0$，将曲线向上平移得到

$$\gamma_C(t) = \gamma(t) + iC.$$

由于所有割缝都在下半平面或实轴上，γ_C 完全落在上半平面且不穿过割缝。上半平面是单连通区域，故 γ_C 可在上半平面内连续收缩为一点。将该同伦向下平移回原位置，即知 γ 在原区域内可缩，因此区域单连通。

20. Other examples of elliptic integrals providing conformal maps from the upper half-plane to rectangles are given below.

(a) The function

$$\int_0^z \frac{d\zeta}{\sqrt{\zeta(\zeta-1)(\zeta-\lambda)}}, \quad \text{with } \lambda \in \mathbb{R} \text{ and } \lambda \neq 1$$

maps the half-plane conformally to a rectangle, one of whose vertices is the image of the point at infinity.

(b) In the case $\lambda = -1$, the image of

$$\int_0^z \frac{d\zeta}{\sqrt{\zeta(\zeta^2-1)}}$$

is a square whose side lengths are $\dfrac{\Gamma^2(1/4)}{2\sqrt{2\pi}}$.

证明 ***(a)*** 由 *Schwara-Christoffel* 积分公式的相关结论，自然有映射得到的图形有四个角，都为 $\dfrac{\pi}{2}$，且将 $0, 1, \lambda, \infty$ 映射为顶点

(b) 当 $\lambda = -1$ 时，矩形的两边长分别为

$$\int_0^1 \frac{d\zeta}{\sqrt{\zeta(1-\zeta^2)}}, \quad \int_1^\infty \frac{d\zeta}{\sqrt{\zeta(\zeta^2-1)}},$$

对第二个式子代换 $\zeta = 1/x$ 有

$$\int_1^\infty \frac{d\zeta}{\sqrt{\zeta(\zeta^2-1)}} = \int_1^0 \frac{-1/x^2}{\sqrt{\frac{1}{x}\left(\frac{1}{x^2}-1\right)}}dx = \int_0^1 \frac{dx}{\sqrt{x(1-x^2)}},$$

故而是正方形. 计算边长

$$\begin{aligned} \text{边长} &= \int_0^1 \frac{dx}{\sqrt{x(1-x^2)}}x = \sin\theta \int_0^{\frac{\pi}{2}} (\sin\theta)^{-1/2} d\theta \\ &= \frac{1}{2}B\left(\frac{1}{4},\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2}\frac{\Gamma(1/4)}{\Gamma(3/4)}, \end{aligned}$$

代入 $\Gamma(1/4)\Gamma(3/4) = \sqrt{2}\pi$ 即有边长 $= \dfrac{\Gamma^2(1/4)}{2\sqrt{2\pi}}$.

24. The elliptic integrals K and K' defined for $0 < k < 1$ by

$$K(k) = \int_0^1 \frac{dx}{((1-x^2)(1-k^2x^2))^{1/2}} \quad \text{and} \quad K'(k) = \int_1^{1/k} \frac{dx}{((x^2-1)(1-k^2x^2))^{1/2}}$$

satisfy various interesting identities. For instance:

(a) Show that if $\tilde{k}^2 = 1-k^2$ and $0 < \tilde{k} < 1$, then

$$K'(k) = K(\tilde{k}).$$

[Hint: Change variables $x = (1-\tilde{k}^2y^2)^{-1/2}$ in the integral defining $K'(k)$.]

(b) Prove that if $\tilde{k}^2 = 1-k^2$, and $0 < \tilde{k} < 1$, then

$$K(k) = \frac{2}{1+\tilde{k}}K\left(\frac{1-\tilde{k}}{1+\tilde{k}}\right).$$

[Hint: Change variables $x = 2t/(1+\tilde{k}+(1-\tilde{k})t^2)$.]

(c) Show that for $0 < k < 1$ one has

$$K(k) = \frac{\pi}{2}F(1/2,1/2,1;k^2),$$

where F the hypergeometric series. [Hint: This follows from the integral representation for F given in Exercise 9, Chapter 6.]

证明 *(a)* 作换元 $x = (1-\tilde{k}^2y^2)^{-1/2}$ 有

$$\begin{aligned} K'(k) &= \int_1^{1/k} \frac{dx}{((x^2-1)(1-k^2x^2))^{1/2}} \\ &= \int_0^1 \frac{-\frac{1}{2}(1-\tilde{k}^2y^2)^{-3/2}(-\tilde{k}^2)2y}{\sqrt{\left(\frac{1}{1-\tilde{k}^2y^2}-1\right)\left(1-\frac{k^2}{1-\tilde{k}^2y^2}\right)}}dy \\ &= \int_0^1 \frac{(1-\tilde{k}^2y^2)^{-1/2}\tilde{k}^2y}{\sqrt{[1-(1-\tilde{k}^2y^2)]\left[(1-\tilde{k}^2y^2)-k^2\right]}}dy \\ &= K(\tilde{k}). \end{aligned}$$

(b) 作换元有

$$
\begin{aligned}
K(k) &= \int_0^1 \frac{dx}{((1-x^2)(1-k^2x^2))^{1/2}} \\
&= \int_0^1 \frac{2\frac{1+\tilde{k}-(1-\tilde{k})t^2}{\left[1+\tilde{k}+(1-\tilde{k})t^2\right]^2}}{\sqrt{\left(1-\frac{4t^2}{\left[1+\tilde{k}+(1-\tilde{k})t^2\right]^2}\right)\left(1-\frac{4k^2t^2}{\left[1+\tilde{k}+(1-\tilde{k})t^2\right]^2}\right)}}dt \\
&= \int_0^1 \frac{2\left[1+\tilde{k}-(1-\tilde{k})t^2\right]}{\sqrt{\left(\left[1+\tilde{k}+(1-\tilde{k})t^2\right]^2-4t^2\right)\left(\left[1+\tilde{k}+(1-\tilde{k})t^2\right]^2-4k^2t^2\right)}}dt \\
&= \int_0^1 \frac{2\left[1+\tilde{k}-(1-\tilde{k})t^2\right]}{\sqrt{\left(\left[1+\tilde{k}+(1-\tilde{k})t^2\right]^2-4t^2\right)\left((1+\tilde{k})-(1-\tilde{k})t^2\right)^2}}dt \\
&= \int_0^1 \frac{2}{\sqrt{\left[1+\tilde{k}+(1-\tilde{k})t^2\right]^2-4t^2}}dt \\
&= \int_0^1 \frac{2}{\sqrt{(1-t^2)\left((1+\tilde{k})^2-(1-\tilde{k})^2t^2\right)}}dt \\
&= \frac{2}{1+\tilde{k}}K\left(\frac{1-\tilde{k}}{1+\tilde{k}}\right),
\end{aligned}
$$

2. The angle between two non-zero complex numbers z and w (taken in that order) is simply the oriented angle, in $(-\pi,\pi]$, that is formed between the two vectors in $\mathbb{R}^2$ corresponding to the points z and w. This oriented angle, say α, is uniquely determined by the two quantities

$$\frac{(z,w)}{|z||w|} \quad \text{and} \quad \frac{(z,-iw)}{|z||w|}$$

which are simply the cosine and sine of α, respectively. Here, the notation $(\cdot,\cdot)$ corresponds to the usual Euclidian inner product in $\mathbb{R}^2$, which in terms of complex numbers takes the form $(z,w)=\mathrm{Re}(z\overline{w})$.

In particular, we may now consider two smooth curves $\gamma:[a,b]\to\mathbb{C}$ and $\eta:[a,b]\to\mathbb{C}$, that intersect at z_0, say $\gamma(t_0)=\eta(t_0)=z_0$, for some $t_0\in(a,b)$. If the quantities $\gamma'(t_0)$ and $\eta'(t_0)$ are non-zero, then they represent the tangents to the curves γ and η at the point z_0, and we say that the two curves intersect at z_0 at the angle formed by the two vectors $\gamma'(t_0)$ and $\eta'(t_0)$.

A holomorphic function f defined near z_0 is said to *preserve angles at* z_0 if for any two smooth curves γ and η intersecting at z_0, the angle formed between the curves γ and η at z_0 equals the angle formed between the curves $f\circ\gamma$ and $f\circ\eta$ at $f(z_0)$. (See Figure 12 for an illustration.) In particular, we assume that the tangents to the curves γ, η, $f\circ\gamma$, and $f\circ\eta$ at the point z_0 and $f(z_0)$ are all non-zero.

(a) Prove that if $f:\Omega\to\mathbb{C}$ is holomorphic, and $f'(z_0)\neq 0$, then f preserves angles at z_0. [Hint: Observe that

$$(f'(z_0)\gamma'(t_0), f'(z_0)\eta'(t_0)) = |f'(z_0)|^2(\gamma'(t_0),\eta'(t_0)).$$

(b) Conversely, prove the following: suppose $f:\Omega\to\mathbb{C}$ is a complex-valued function, that is real-differentiable at $z_0\in\Omega$, and $J_f(z_0)\neq 0$. If f preserves angles at z_0, then f is holomorphic at z_0 with $f'(z_0)\neq 0$.

证明 ***(a)*** 映射后曲线切向量为

$$(f\circ\gamma)'(t_0) = f'(z_0)\gamma'(t_0),\quad (f\circ\eta)'(t_0) = f'(z_0)\eta'(t_0).$$

计算内积:

$$(f'\gamma', f'\eta') = \mathrm{Re}\left(f'\gamma'\cdot\overline{f'\eta'}\right) = |f'(z_0)|^2(\gamma',\eta').$$

同理

$$(f'\gamma', -if'\eta') = |f'(z_0)|^2(\gamma', -i\eta').$$

因此切向量夹角的余弦与正弦均不变，故 f 在 z_0 处保角。

(b)

设 $f(z)=u(x,y)+iv(x,y)$，$z=x+iy$，由 $f\in C^1(\Omega)$，f 在 $z_0=(x_0,y_0)$ 处的 *Jacobi* 矩阵为

$$J=\begin{pmatrix}\frac{\partial u}{\partial x}(z_0) & \frac{\partial u}{\partial y}(z_0)\\ \frac{\partial v}{\partial x}(z_0) & \frac{\partial v}{\partial y}(z_0)\end{pmatrix}.$$

f 在 z_0 处保角，等价于：对任意二维向量 $\boldsymbol{a},\boldsymbol{b}\in\mathbb{R}^2$，存在固定正常数 $k>0$，使得

$$\langle\boldsymbol{a},\boldsymbol{b}\rangle = k\langle J\boldsymbol{a}, J\boldsymbol{b}\rangle,$$

其中 $\langle\cdot,\cdot\rangle$ 为 $\mathbb{R}^2$ 上的标准欧氏内积。由内积的矩阵表示 $\langle\boldsymbol{a},\boldsymbol{b}\rangle=\boldsymbol{a}^TE\boldsymbol{b}$（$E$ 为单位矩阵），代入得

$$\boldsymbol{a}^TE\boldsymbol{b} = k\boldsymbol{a}^TJ^TJ\boldsymbol{b},\quad \forall\boldsymbol{a},\boldsymbol{b},$$

因此矩阵等式

$$E=kJ^TJ \quad\Longleftrightarrow\quad J^TJ=\frac{1}{k}E.$$

计算 J^TJ:

$$J^TJ=\begin{pmatrix}\left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial v}{\partial x}\right)^2 & \frac{\partial u}{\partial x}\frac{\partial u}{\partial y}+\frac{\partial v}{\partial x}\frac{\partial v}{\partial y}\\ \frac{\partial u}{\partial x}\frac{\partial u}{\partial y}+\frac{\partial v}{\partial x}\frac{\partial v}{\partial y} & \left(\frac{\partial u}{\partial y}\right)^2+\left(\frac{\partial v}{\partial y}\right)^2\end{pmatrix}.$$

结合 $J^TJ=\frac{1}{k}E$，得到方程组:

$$\begin{cases}\left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial v}{\partial x}\right)^2=\left(\frac{\partial u}{\partial y}\right)^2+\left(\frac{\partial v}{\partial y}\right)^2=\frac{1}{k},\\ \dfrac{\partial u}{\partial x}\dfrac{\partial u}{\partial y}+\dfrac{\partial v}{\partial x}\dfrac{\partial v}{\partial y}=0.\end{cases}$$

解此方程组，仅存在两种可能：

$$\text{要么}\begin{cases}\dfrac{\partial u}{\partial x}=\dfrac{\partial v}{\partial y},\\ \dfrac{\partial u}{\partial y}=-\dfrac{\partial v}{\partial x},\end{cases}\qquad \text{要么}\begin{cases}\dfrac{\partial u}{\partial x}=-\dfrac{\partial v}{\partial y},\\ \dfrac{\partial u}{\partial y}=\dfrac{\partial v}{\partial x}.\end{cases}$$

引入复偏导数

$$\frac{\partial f}{\partial z}=\frac{1}{2}\left(\frac{\partial f}{\partial x}-i\frac{\partial f}{\partial y}\right),\quad \frac{\partial f}{\partial \bar{z}}=\frac{1}{2}\left(\frac{\partial f}{\partial x}+i\frac{\partial f}{\partial y}\right).$$

若满足第一个方程，则 $\dfrac{\partial f}{\partial \bar{z}}=0$，即 f 在 z_0 处全纯，且 $|f'(z_0)|=\left|\dfrac{\partial f}{\partial z}\right|=\dfrac{1}{\sqrt{k}}\neq 0$；

若满足第二个方程，则 $\dfrac{\partial f}{\partial z}=0$，$f$ 为反全纯函数，此时会将逆时针夹角变为顺时针夹角，与题目中保定向保角的要求矛盾，故排除。

因此仅可能 f 在 z_0 处全纯，且 $f'(z_0)\neq 0$。

3.* The Schwarz-Pick lemma (see Exercise 13) is the infinitesimal version of an important observation in complex analysis and geometry.

For complex numbers $w\in\mathbb{C}$ and $z\in\mathbb{D}$ we define the *hyperbolic length* of w at z by

$$\|w\|_z=\frac{|w|}{1-|z|^2},$$

where $|w|$ and $|z|$ denote the usual absolute values. This length is sometimes referred to as the *Poincaré metric*, and as a Riemann metric it is written as

$$ds^2=\frac{|dz|^2}{(1-|z|^2)^2}.$$

The idea is to think of w as a vector lying in the tangent space at z. Observe that for a fixed w, its hyperbolic length grows to infinity as z approaches the boundary of the disc. We pass from the infinitesimal hyperbolic length of tangent vectors to the global hyperbolic distance between two points by integration.

(a) Given two complex numbers z_1 and z_2 in the disc, we define the hyperbolic distance between them by

$$d(z_1,z_2)=\inf_{\gamma}\int_0^1\|\gamma'(t)\|_{\gamma(t)}dt,$$

where the infimum is taken over all smooth curves $\gamma:[0,1]\to\mathbb{D}$ joining z_1 and z_2. Use the Schwarz-Pick lemma to prove that if $f:\mathbb{D}\to\mathbb{D}$ is holomorphic, then

$$d(f(z_1),f(z_2))\leq d(z_1,z_2)\quad\text{for any } z_1,z_2\in\mathbb{D}.$$

In other words, holomorphic functions are distance-decreasing in the hyperbolic metric.

(b) Prove that automorphisms of the unit disc preserve the hyperbolic distance, namely

$$d(\varphi(z_1),\varphi(z_2))=d(z_1,z_2),\quad\text{for any } z_1,z_2\in\mathbb{D}$$

and any automorphism φ. Conversely, if $\varphi:\mathbb{D}\to\mathbb{D}$ preserves the hyperbolic distance, then either φ or $\overline{\varphi}$ is an automorphism of $\mathbb{D}$.

(c) Given two points $z_1, z_2 \in \mathbb{D}$, show that there exists an automorphism φ such that $\varphi(z_1) = 0$ and $\varphi(z_2) = s$ for some s on the segment $[0, 1)$ on the real line.

(d) Prove that the hyperbolic distance between 0 and $s \in [0, 1)$ is

$$d(0, s) = \frac{1}{2} \log \frac{1+s}{1-s}.$$

(e) Find a formula for the hyperbolic distance between any two points in the unit disc.

证明 ***(a)*** 由 *Schwarz–Pick* 引理，对任意 $z \in \mathbb{D}$ 有

$$\frac{|f'(z)|}{1-|f(z)|^2} \leq \frac{1}{1-|z|^2}.$$

对任意连接 z_1, z_2 的曲线 γ，其像曲线 $f \circ \gamma$ 的长度满足

$$\int_0^1 \|(f\circ\gamma)'(t)\|_{f(\gamma(t))} dt = \int_0^1 \frac{|f'(\gamma(t))\gamma'(t)|}{1-|f(\gamma(t))|^2} dt \leq \int_0^1 \frac{|\gamma'(t)|}{1-|\gamma(t)|^2} dt.$$

对所有曲线取下确界，即得

$$d(f(z_1), f(z_2)) \leq d(z_1, z_2).$$

(b) φ 与逆映射 φ^{-1} 都是 $\mathbb{D} \to \mathbb{D}$ 的全纯映射，由 *(a)*：

$$d(\varphi(z_1), \varphi(z_2)) \leq d(z_1, z_2), \quad d(z_1, z_2) = d(\varphi^{-1}(\varphi z_1), \varphi^{-1}(\varphi z_2)) \leq d(\varphi z_1, \varphi z_2).$$

故等号成立，φ 保持双曲距离。

(c) 令

$$\varphi_1(z) = \frac{z - z_1}{1 - \overline{z_1} z},$$

则 $\varphi_1(z_1) = 0$。记 $w = \varphi_1(z_2)$，再取旋转

$$\varphi_2(z) = e^{-i \arg w} z,$$

则 $\varphi = \varphi_2 \circ \varphi_1$ 即为所求。

(d) 连接 0 与 s 的最短曲线为实直线段，直接计算：

$$d(0, s) = \int_0^s \frac{dx}{1-x^2} = \frac{1}{2} \left(\ln(1+x) - \ln(1-x) \right) \Big|_0^s = \frac{1}{2} \ln \frac{1+s}{1-s}.$$

(e)

利用 *(c)*，取自同构 φ 使得 $\varphi(z_1) = 0$，$\varphi(z_2) = s$，则

$$s = \left| \frac{z_2 - z_1}{1 - \overline{z_1} z_2} \right|.$$

由距离不变性及 *(d)*，得

$$d(z_1, z_2) = d(0, s) = \frac{1}{2} \ln \frac{1 + \left| \dfrac{z_2 - z_1}{1 - \overline{z_1} z_2} \right|}{1 - \left| \dfrac{z_2 - z_1}{1 - \overline{z_1} z_2} \right|}.$$

4.* Consider the group of matrices of the form

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

that satisfy the following conditions:

(i) a, b, c, and $d \in \mathbb{C}$,

(ii) the determinant of M is equal to 1,

(iii) the matrix M preserves the following hermitian form on $\mathbb{C}^2 \times \mathbb{C}^2$:

$$\langle Z, W \rangle = z_1 \overline{w}_1 - z_2 \overline{w}_2,$$

where $Z = (z_1, z_2)$ and $W = (w_1, w_2)$. In other words, for all $Z, W \in \mathbb{C}^2$

$$\langle MZ, MW \rangle = \langle Z, W \rangle.$$

This group of matrices is denoted by $\mathrm{SU}(1,1)$.

(a) Prove that all matrices in $\mathrm{SU}(1,1)$ are of the form

$$\begin{pmatrix} a & b \\ \overline{b} & \overline{a} \end{pmatrix},$$

where $|a|^2 - |b|^2 = 1$. To do so, consider the matrix

$$J = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

and observe that $\langle Z, W \rangle = {}^tWJZ$, where ${}^tW$ denotes the conjugate transpose of W.

(b) To every matrix in $\mathrm{SU}(1,1)$ we can associate a fractional linear transformation

$$\frac{az + b}{cz + d}.$$

Prove that the group $\mathrm{SU}(1,1)/\{\pm 1\}$ is isomorphic to the group of automorphisms of the disc. [Hint: Use the following association.]

$$e^{2i\theta} \frac{z - \alpha}{1 - \overline{\alpha} z} \to \begin{pmatrix} \dfrac{e^{i\theta}}{\sqrt{1 - |\alpha|^2}} & -\dfrac{\alpha e^{i\theta}}{\sqrt{1 - |\alpha|^2}} \\ -\dfrac{\overline{\alpha} e^{-i\theta}}{\sqrt{1 - |\alpha|^2}} & \dfrac{e^{-i\theta}}{\sqrt{1 - |\alpha|^2}} \end{pmatrix}.$$

证明 ***(a)*** 由条件 $\langle MZ, MW \rangle = \langle Z, W \rangle$，得

$$M^* J M = J, \quad J = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

结合 $\det M = 1$，展开矩阵等式可得 $c = \overline{b}$，$d = \overline{a}$ 且 $|a|^2 - |b|^2 = 1$。

(b) 单位圆盘自同构形如

$$\varphi(z) = e^{2i\theta} \frac{z - \alpha}{1 - \overline{\alpha} z}.$$

对应 $\mathrm{SU}(1,1)$ 中矩阵

$$M = \frac{1}{\sqrt{1-|\alpha|^2}} \begin{pmatrix} e^{i\theta} & -\alpha e^{i\theta} \\ -\overline{\alpha} e^{-i\theta} & e^{-i\theta} \end{pmatrix}.$$

M 与 $-M$ 对应同一个分式线性变换，故

$$\mathrm{SU}(1,1)/\{\pm I\} \cong \mathrm{Aut}(\mathbb{D}).$$

8.* Let f be an injective holomorphic function in the unit disc, with $f(0) = 0$ and $f'(0) = 1$. If we write $f(z) = z + a_2 z^2 + a_3 z^3 + \ldots$, then Problem 1 in Chapter 3 shows that $|a_2| \leq 2$. Bieberbach conjectured that in fact $|a_n| \leq n$ for all $n \geq 2$; this was proved by deBranges. This problem outlines an argument to prove the conjecture under the additional assumption that the coefficients a_n are real.

(a) Let $z = re^{i\theta}$ with $0 < r < 1$, and show that if $v(r,\theta)$ denotes the imaginary part of $f(re^{i\theta})$, then

$$a_n r^n = \frac{2}{\pi} \int_0^\pi v(r,\theta) \sin n\theta d\theta.$$

(b) Show that for $0 \leq \theta \leq \pi$ and $n = 1, 2, \ldots$ we have $|\sin n\theta| \leq n \sin\theta$.

(c) Use the fact that $a_n \in \mathbb{R}$ to show that $f(\mathbb{D})$ is symmetric with respect to the real axis, and use this fact to show that f maps the upper half-disc into either the upper or lower part of $f(\mathbb{D})$.

(d) Show that for r small,

$$v(r,\theta) = r \sin\theta[1 + O(r)],$$

and use the previous part to conclude that $v(r,\theta)\sin\theta \geq 0$ for all $0 < r < 1$ and $0 \leq \theta \leq \pi$.

(e) Prove that $|a_n r^n| \leq nr$, and let $r \to 1$ to conclude that $|a_n| \leq n$.

(f) Check that the function $f(z) = z/(1-z)^2$ satisfies all the hypotheses and that $|a_n| = n$ for all n.

证明 ***(a)*** 由展开式

$$f(re^{i\theta}) = \sum_{k=1}^{\infty} a_k r^k e^{ik\theta},$$

取虚部得

$$v(r,\theta) = \sum_{k=1}^{\infty} a_k r^k \sin k\theta.$$

两边同乘 $\sin n\theta$ 并在 $[0,\pi]$ 上积分，由三角函数系的正交性即得

$$a_n r^n = \frac{2}{\pi} \int_0^\pi v(r,\theta) \sin n\theta \, d\theta.$$

(b) 利用恒等式

$$\sin n\theta = \sin\theta\big(1 + 2\cos 2\theta + 2\cos 4\theta + \cdots + 2\cos(n-1)\theta\big).$$

括号内绝对值不超过 n，故 $|\sin n\theta| \leq n \sin\theta$。

(c) 由 $a_n \in \mathbb{R}$，有

$$\overline{f(z)} = f(\overline{z}),$$

故像集关于实轴对称。

上半圆盘在 f 下的像若穿过实轴，则考虑其共轭，结合对称性，与但同态矛盾，从而上半圆盘在 f 下的像不穿过实轴，只能整体落在 $f(\mathbb{D})$ 的上半部分或下半部分。

(d) 展开得

$$v(r,\theta) = r\sin\theta + a_2 r^2 \sin 2\theta + \cdots = r\sin\theta\big(1 + O(r)\big).$$

当 r 充分小时，v 与 $\sin\theta$ 同号。由 *(c)* 知 v 在 $0 < \theta < \pi$ 内不变号，故对所有 $0 < r < 1$ 都有 $v(r,\theta)\sin\theta \geq 0$。

(e) 由 *(a)(b)(d)*，

$$|a_n| r^n = \frac{2}{\pi}\left|\int_0^\pi v \sin n\theta d\theta\right| \leq \frac{2}{\pi}\int_0^\pi v \cdot n\sin\theta d\theta = n \cdot \frac{2}{\pi}\int_0^\pi v\sin\theta d\theta.$$

而

$$\frac{2}{\pi}\int_0^\pi v\sin\theta d\theta = a_1 r = r.$$

因此 $|a_n| r^n \leq nr$，即 $|a_n| \leq nr^{1-n}$。令 $r \to 1^-$，得 $|a_n| \leq n$。

(f) 对 $f(z) = \dfrac{z}{(1-z)^2}$ 作幂级数展开：

$$f(z) = z\sum_{n=1}^{\infty} n z^{n-1} = \sum_{n=1}^{\infty} n z^n.$$

显然 $f(0) = 0$，$f'(0) = 1$，且 f 在单位圆盘内单叶，系数 $a_n = n$，达到上界。